

Degree: B.Tech, Stream: ALL (Group-A)

Term – I, First Semester Examination, August - 2025

Subject Code: BSCBE104

Subject Name: Biology for Engineers

Full Marks: 30

Duration: 1 Hour

Part - A

Answer any 5 out of 6 questions

Each question carries 2 marks (2 × 5)

1. Demonstrate how the second law of thermodynamics is reflected in biological systems. 2
2. DNA is more stable than RNA. Elaborate on the statement 2
3. Differentiate DNA polymerase and RNA polymerase. 2
4. Discuss the function of DNA Pol I and Topoisomerase I. 2
5. Discuss about two fields in biological science with strong engineering job prospects. 2
6. What are the implementations of 3D-Bioprinting? 2

Part - B

Answer any 2 out of 3 questions

Each question carries 5 marks (5 × 2)

7. Differentiate between exergonic and endergonic reactions. Give examples of each. 5
8. DNA replication is semiconservative in nature-Justify. 5
9. Illustrate the phenomenon of "Brownian Motion" and its aspects in scientific fields. 5

Part - C

Answer any 1 out of 2 questions

Each question carries 10 marks (10 × 1)

10. For the reaction:
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ occurring at 25°C, the standard Gibbs free energy change is -33.0 kJ/mol. 10

The partial pressures of the gases at a certain point in the reaction are:

$$\Delta G = -nRT$$

- $P_{N_2}=0.5 \text{ atmP}$
- $P_{H_2}=1.0 \text{ atmP}$
- $P_{NH_3}=4.0 \text{ atmP}$

- Calculate the reaction quotient Q .
- Calculate the change in free energy under these non-standard conditions.
- Determine whether the product or the reactants are favoured in the reaction under this condition.

11. Differentiate between the three forms of the DNA.

10

--End--